MachiNest

LEAN MANUFACTURING BASICS

The 7 Wastes in Manufacturing

Learn to see waste on the shop floor, with CNC and machine-shop examples and simple ways to reduce each one.

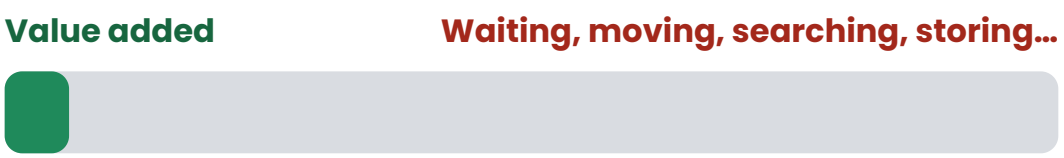
Transportation | Inventory | Motion | Waiting | Over-production | Over-processing | Defects



INTRODUCTION

What is waste (Muda)?

Waste is any activity that uses time, money or effort but does not add value for the customer. The customer pays for a part with the right shape, size and finish, not for the time it spends waiting, moving or being searched for.



In many shops, only a small share of a part's total lead time is actual machining. The rest is waiting, moving and storing.

Three kinds of activity

Value added

Changes the part the way the customer wants. Example: turning, milling, grinding, required heat treatment.

Necessary, no value

Needed today but does not change the part. Example: loading the part, required inspection, setup.

Pure waste

Can be removed. Example: searching for a gauge, waiting for material, rework.

Remember the 7 wastes as TIMWOOD

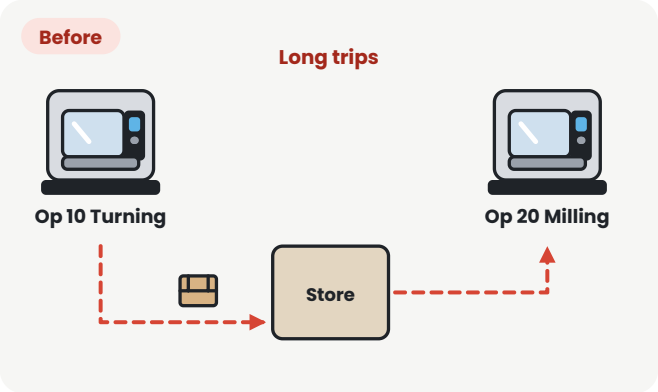
	Waste	Hindi	Machine-shop example
T	**Transportation**	परिवहन	Parts sent to a store between two operations
I	**Inventory**	ज़्यादा माल	Bins of WIP waiting for days
M	**Motion**	बेकार हलचल	Walking to the tool crib for inserts
W	**Waiting**	इंतज़ार	Machine idle for first-off approval
O	**Over-production**	ज़्यादा उत्पादन	Extra parts made because setup is done
O	**Over-processing**	ज़्यादा काम	Polishing a surface the drawing does not need
D	**Defects**	ख़राबी	Rework and scrap from wrong size

WASTE 1 · T

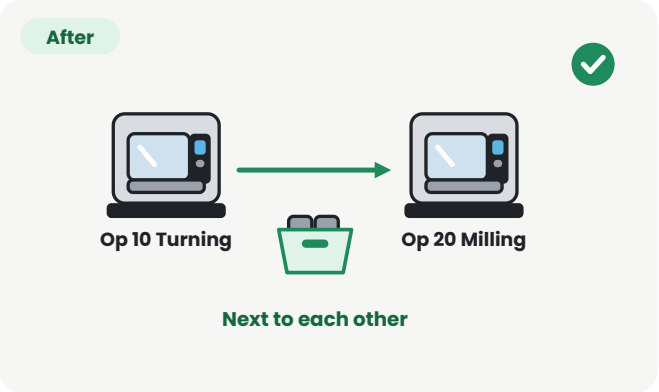
1 Transportation

Moving material, parts or tools more than needed. Moving adds cost and risk of damage, but no value.

Ask yourself How many metres does one part travel from raw material to dispatch?



Parts travel from one machine to the store and back to the next machine: long trips, more handling, more dents.



The next operation is placed close by. Parts move a few steps in a small bin.

Machine-shop examples

- Parts taken to a central store between turning and milling
- Many forklift trips a shift to fetch bar stock
- Parts carried to a far inspection room for every check
- Sending parts to a vendor for a small operation that could be done in-house

How to reduce it

1. Draw a spaghetti diagram: plot how one part moves on the shop layout.
2. Place machines in the order of the process (cell layout).
3. Keep the input bin of the next operation next to the previous machine.
4. Use trolleys on fixed routes instead of many hand-carries.
5. Bring gauges to the machine instead of taking parts to the lab.

Common cause

Machines grouped by type (all lathes together, all VMCs together) instead of by process flow.

Quick win

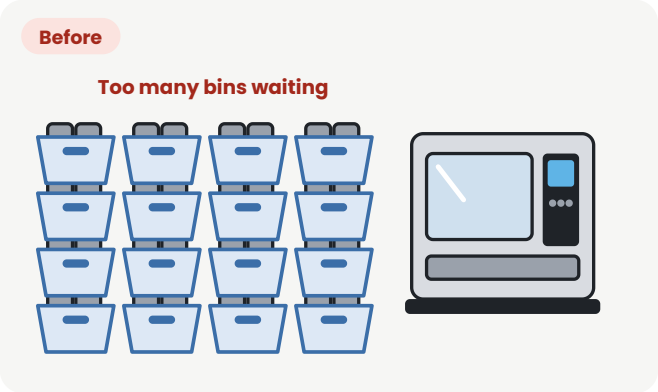
Draw the real path of one part number on the shop layout. Most shops are surprised by the distance.

WASTE 2 · I

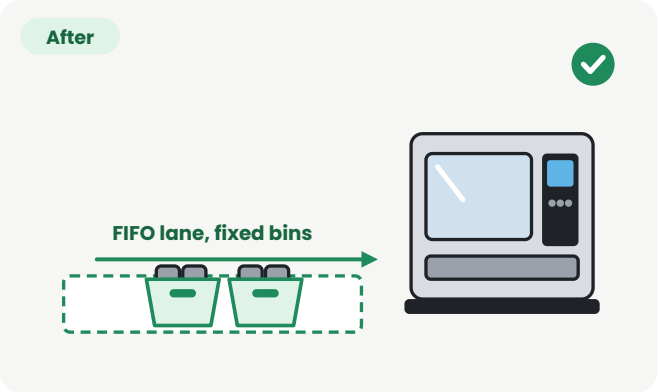
2 Inventory

More raw material, WIP or finished parts than needed right now. Stock hides problems and locks up money.

Ask yourself How many days of work are waiting in front of this machine?



Bins pile up in front of the machine. Old parts get mixed, rusted or lost.



A small FIFO lane with a fixed number of bins: first in, first out.

Machine-shop examples

- Bins of semi-finished parts waiting days between operations
- Bar stock bought in bulk lying for months
- Finished parts in stock because they were made early
- Old revision parts that no one will buy

How to reduce it

1. Set a maximum number of bins (WIP limit) between operations.
2. Mark FIFO lanes on the floor.
3. Make smaller batches that match the next process.
4. Order raw material closer to need, with min–max levels.
5. Review slow-moving stock every month.

Common cause

Large batches and long setups, so everyone makes more "to be safe".

Quick win

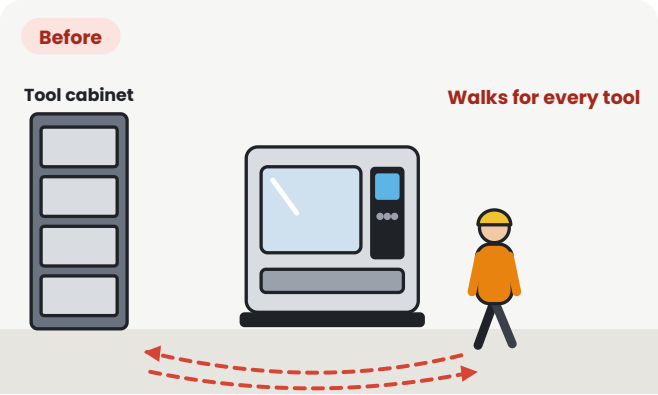
Count the bins waiting between two machines today. Mark a lane that holds only half of that.

WASTE 3 · M

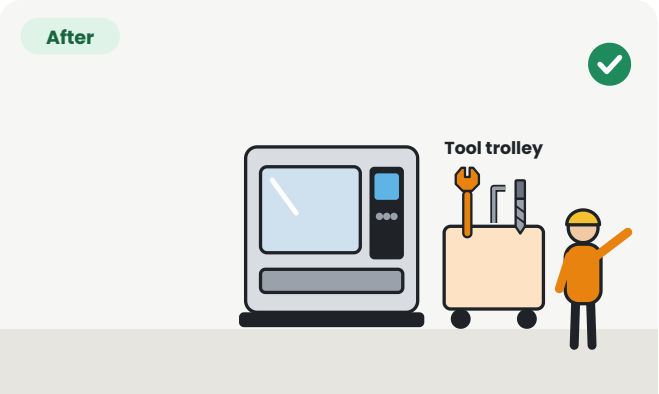
3 Motion

Unnecessary movement of people: walking, bending, reaching and searching. It tires people and wastes time.

Ask yourself How many times did the operator leave the machine in the last hour?



The operator walks to a far cabinet for every tool or gauge.



A tool trolley at the machine with everything needed for the job.

Machine-shop examples

- Walking to the tool crib for inserts many times a shift
- Bending to lift heavy bars or parts from the floor
- Searching for the right Allen key or gauge
- Climbing to reach a tool on a high shelf

How to reduce it

1. Watch one operator for 30 minutes and note every walk and search.
2. Keep tools and gauges within arm's reach (5S: Set in order).
3. Use tool trolleys and setup kits for each part number.
4. Raise bins to waist height with stands or tilted racks.
5. Use lifting aids for heavy parts.

Common cause

Tools kept in a central crib "to control them", far from where the work happens.

Quick win

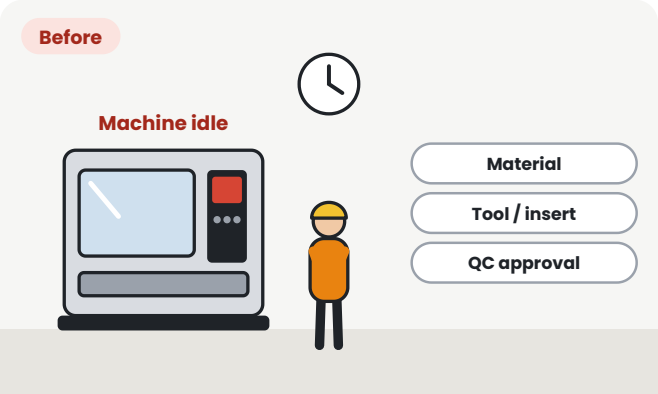
Put a kit of the five most-used tools at each machine and count the walks saved in one shift.

WASTE 4 · W

4 Waiting

People or machines standing idle while they wait for material, tools, programs, approvals or repair.

Ask yourself How many minutes did this machine stand idle today, and why?



The machine is stopped and the operator waits for material, a tool or QC approval.



Material, drawing and preset tools are ready before the job starts.

Machine-shop examples

- Waiting for first-off approval from quality
- Waiting for the setter or the program
- Machine idle because raw material is not cut
- Waiting for a crane or forklift

How to reduce it

1. Prepare the next job's material, tools and program while the current job runs.
2. Preset tools offline, away from the machine.
3. Let trained operators do first-off checks with clear gauges.
4. Record the reason for every machine stoppage each day.
5. Plan maintenance at fixed times.

Common cause

Setup work starts only after the machine stops, instead of while it is still running.

Quick win

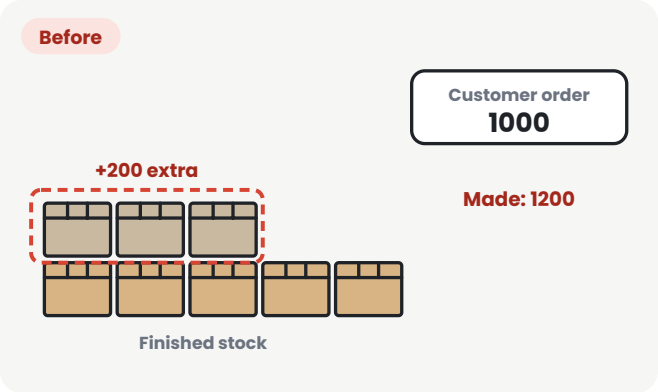
Make a "next job" trolley: material, drawing, program and tools ready before the changeover.

WASTE 5 · O

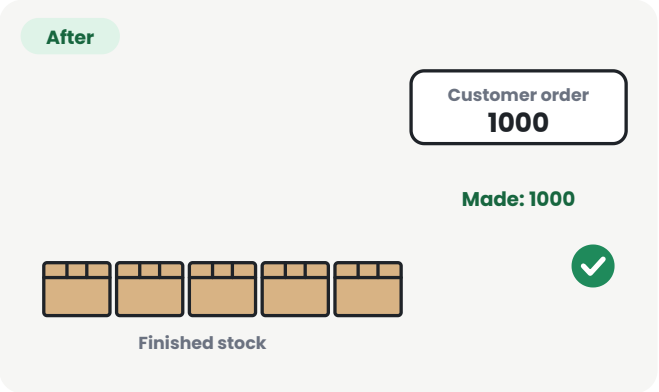
5 Over-production

Making more, or earlier, than the customer needs. It is the worst waste because it creates the others: stock, handling and storage.

Ask yourself Is there a customer order for every part we are making today?



The order is 1000, but 1200 are made "because the setup is done". 200 go to stock.



Make what is ordered, when it is needed.

Machine-shop examples

- Running extra parts because the setup is already done
- Making next month's order early to keep the machine busy
- Producing before the drawing revision is confirmed
- Running the full batch before the first-off is approved

How to reduce it

1. Produce to the customer schedule, not to keep machines busy.
2. Reduce setup time so smaller batches make sense (SMED).
3. Use a board showing plan vs actual every hour.
4. Stop when the order quantity is reached.
5. Agree a small, fixed allowance for setup scrap.

Common cause

Success measured by machine hours or pieces made, instead of orders delivered on time.

Quick win

Write the order quantity on the job card and on the machine board, and stop at that number.

WASTE 6 · O

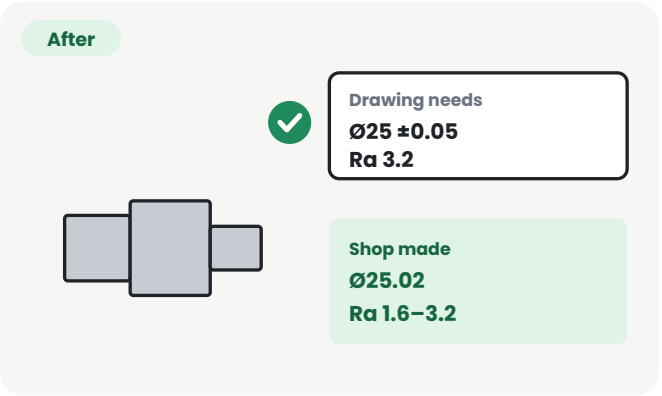
6 Over-processing

Doing more work than the customer needs: tighter tolerance, finer finish, extra operations or extra checks.

Ask yourself Does the drawing really ask for this pass, finish or check?



The drawing needs ±0.05 and Ra 3.2, but the part is made to ±0.005 and Ra 0.4, plus polishing.



The part is made to the drawing: the right size and finish, nothing extra.

Machine-shop examples

- Finishing passes or polishing that the drawing does not need
- Hand deburring that the program could chamfer in-cycle
- 100% inspection of a stable, non-critical size
- Extra roughing passes in a program that was never optimised

How to reduce it

1. Read the drawing carefully: tolerance, finish and notes.
2. Ask the customer when a requirement is unclear.
3. Optimise the program: fewer passes, combined operations, chamfer in-cycle.
4. Match inspection frequency to how critical and stable the feature is.
5. Review old programs when tools or machines change.

Common cause

"Better safe than sorry" habits, and not knowing what the customer really needs.

Quick win

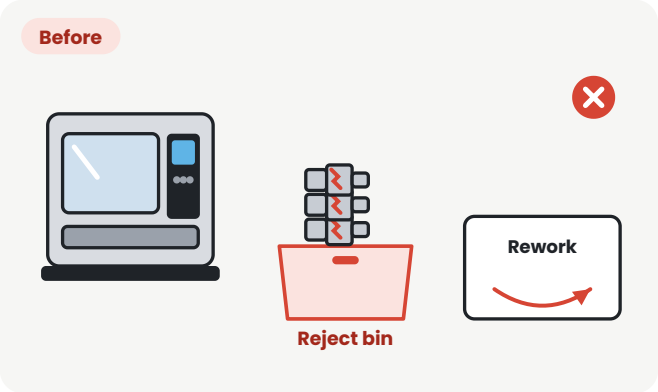
Pick one high-volume part and remove one pass or operation that adds nothing.

WASTE 7 · D

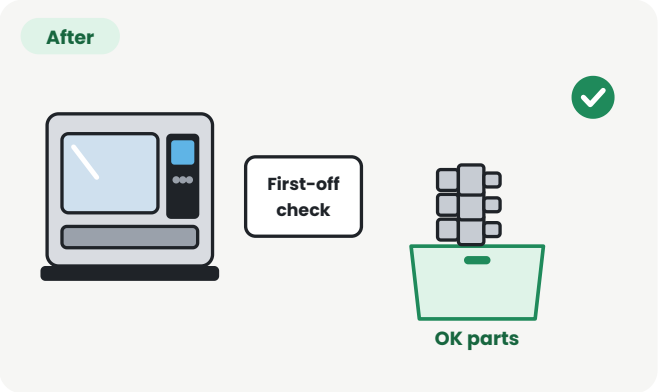
7 Defects

Parts that are wrong and need rework or scrap. Every defect wastes material, machine time, labour and the customer's trust.

Ask yourself Which defect came back most often this month?



Bad parts go to the reject bin or come back for rework.



First-off and in-process checks catch problems early.

Machine-shop examples

- Size out of tolerance from tool wear or an offset error
- Wrong drawing revision used
- Burrs, dents and rust from handling and storage
- Thread not gauging because of a wrong pitch in the program

How to reduce it

1. Do a first-off check after every setup and tool change.
2. Keep go/no-go gauges at the machine.
3. Track tool life and change inserts before they fail.
4. Find the root cause of repeat defects with "5 Why".
5. Mistake-proof the process (Poka-yoke).

Common cause

No standard first-off and in-process checks, so problems are found only at final inspection.

Quick win

Start a reject board: part, defect, count and cause every day. The top repeat defect is your first kaizen.

BONUS

8 The 8th waste: unused talent

Not using people's ideas, skills and experience. Operators often know the quickest fix, but no one asks them.

Example: an operator knows a better way to clamp a part that would save 2 minutes a cycle, but there is no way to share it.

1. Hold a 5-minute meeting at the start of each shift.
2. Put up a simple idea board at each area.
3. Try good ideas quickly and thank people in public.

Waste walk checklist

Walk the shop floor for 20 minutes. Tick each waste you see and write where.

Waste	What to look for	Seen?	Where / notes
Transportation	Long part movements, forklift trips, parts going to a store and back		
Inventory	Bins piling up, old stock, parts without a tag or date		
Motion	Walking for tools, bending, searching, reaching high		
Waiting	Idle machines or people, waiting for material, setter or QC		
Over-production	Parts made beyond the order, or long before they are due		
Over-processing	Extra passes, finish or checks beyond the drawing		
Defects	Reject bins, rework tables, repeat customer complaints		
Unused talent	Ideas not asked for, skills not used		